MVAdapt: Zero-Shot Multi-Vehicle Adaptation for End-to-End Autonomous Driving

Abstract— **One of the recent focuses of AI research is End-to-End Autonomous Driving, as it is gaining interest due to its promising potential. However, it has an unexplored limitation: driving performance decreases when an E2E driving AI model is deployed in vehicles that differ from those used in training, which we denote as *vehicle domain gap*. To address this problem, this paper introduces the framework *MVAdapt*, a novel foundation model for effective adaptation to various vehicle archetypes. The main contribution is a low-level multi-head transformer encoder that generates physics-integrated scene feature embeddings. By deriving physical properties of the target vehicle, MVAdapt dynamically decodes target-vehicle-specific waypoints with real-time performance. Even in an unseen vehicle that has extreme physical differences, MVAdapt proves its few-shot adaptation ability, as it quickly closes the performance gap by leveraging a tiny dataset and fine-tuning with it. The evaluation experiments performed with the CARLA Leaderboard 1.0 benchmark demonstrate that MVAdapt outperforms state-of-the-art performance across various metrics. Our work presents a practical method for achieving robust cross-vehicle generalization, pioneering the way for more scalable and adaptable autonomous driving AI. Moreover, we propose a new research area for autonomous driving, *vehicle-domain adaptation*.**

I. Introduction

The conventional autonomous driving divides the pipeline into several modules: perception, localization, planning, and control, for instance. In contrast, the End-to-end (E2E) autonomous driving aims to learn a single AI model that directly maps raw sensor data inputs to vehicle control commands. The approach has an advantage over the conventional way for preventing error propagation through modules by optimizing the entire system jointly. In this regard, it solves the primary hurdle of traditional modular pipelines. [1]–[5] However, generalization capability to untrained domains remains a significant barrier. Recent works have focused on environment domain adaptations, such as sim-to-real adaptations [6]–[9], climate adaptations [10], [11], and region adaptations [12]–[15]. In this paper, we introduce a novel domain adaptation paradigm, which we term *vehicle-domain adaptation*, as [16] suggests its necessity. This research point of view has been overlooked because current E2E models treat vehicle dynamics as part of the hidden feature to be approximated, rather than as an explicit factor to be learned.

E2E models learn an implicit mapping from perception to control, which inherently embeds vehicle-specific dynamics. For instance, an E2E model trained on a lightweight sedan encodes a sedan-specific driving maneuver by learning the expert actions for the vehicle. When it is naively transferred to a big and heavy SUV, the same driving outputs with the same sensor inputs cause critical dangers. First, an SUV and a sedan will exhibit distinct dynamic responses even when subjected to identical control inputs. Second, trajectories considered safe for the sedan may be infeasible or hazardous for the SUV. Lastly, the model may generate a command that is kinematically infeasible for the SUV to execute.

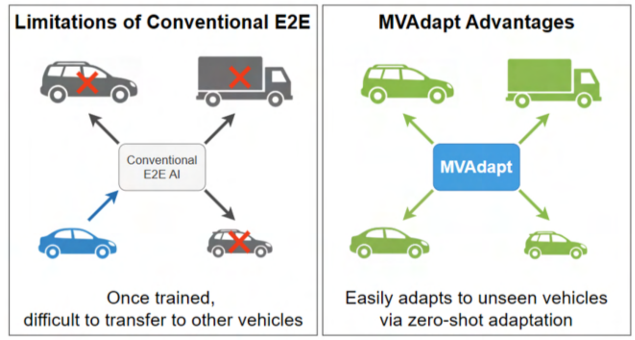

Fig. 1: *Advantages of MVAdapt*: Conventional E2E models lack transferability across vehicles (left), while MVAdapt enables zero-shot adaptation to unseen vehicle types (right).

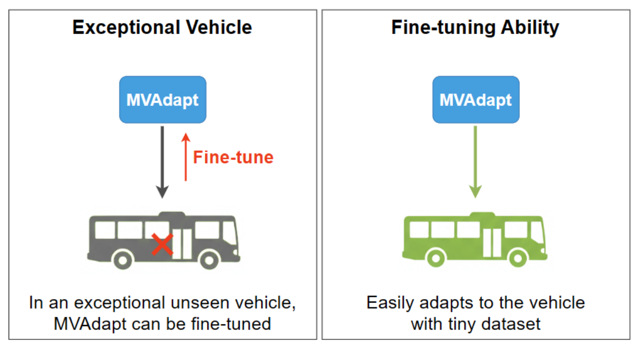

Fig. 2: *Few-shot Adaptation*: Even if MVAdapt is not able to adapt to an exceptional unseen vehicle in zero-shot manner (left), it shows fine-tuning ability with a minimal dataset (right).

Especially, the process of data collection and model re-training for each target vehicle is prohibitively resource-intensive, hindering the widespread deployment of E2E models. Therefore, to achieve effective vehicle-domain adaptation, the model must internalize the physical attributes of a vehicle and the requisite adjustments in its driving policy.

To address this vehicle domain gap, we propose **MVAdapt**, a framework that integrates across-vehicle autonomous driving.

Our contributions are the following three:

- We demonstrate and analyze the *vehicle-domain gap* as a critical challenge for E2E autonomous driving. Practically, we propose a training strategy of integrating the physical properties of vehicle platforms.
- We propose **MVAdapt**, a novel architecture that conditions the driving outputs with physical parameters using a cross-attention transformer algorithm. This foundation model carries significant zero-shot adaptation performance in unseen vehicles, while also possessing a robust capacity for few-shot adaptation in more challenging scenarios.
- We demonstrate experiments in the CARLA Leaderboard 1.0 Benchmark simulator to show that MVAdapt outperforms strong cross-platform baselines from the robotics domain, yielding up to 135.22% improvement in the Driving Score. It achieves state-of-the-art performance in vehicle-domain adaptation.

The remainder of the paper is structured as follows: Section II outlines related previous works. Afterward, Section III derives the framework MVAdapt. The validation experiments are presented in Section IV. After discussing the results in Section V, Section VI summarizes the findings and suggests directions for future work.

II. Related Works

A. End-to-End Autonomous Driving

The field has evolved from early CNN-based models like PilotNet [17], [18] to more sophisticated architectures. More recently, transformer-based models such as TransFuser [19], [20] and InterFuser [21] have set new performance benchmarks by effectively fusing multi-sensor data. Despite these advances, most research implicitly assumes a fixed vehicle embodiment, and the problem of generalization across different vehicle dynamics remains largely unaddressed.

B. Domain Adaptation in End-to-End Autonomous Driving

A significant body of research has focused on bridging other domain gaps, such as the sim-to-real gap [6]–[9] or adapting to varied environmental conditions like climate [10], [11] and region [12], [13]. To support such domain shift problems, a parallel line of work has conducted dedicated datasets [22], [23]. These methods typically focus on achieving visual feature invariance and do not provide mechanisms to adapt the control policy to changes in the agent's underlying physical properties.

C. Physics-Informed Learning in End-to-End Autonomous Driving

Another line of research seeks to integrate physical knowledge into learning-based models. This includes hybrid approaches that couple deep learning perception modules with classical controllers like MPC [24]–[26]. More autonomous driving works based on Physics-Informed Neural Networks (PINNs) exist, such as Deep Dynamics [27] and Fusion-Assurance [28], which learn to generate physics-constrained maneuvers. In contrast, MVAdapt's objective is to condition the policy on known a priori physical parameters to enable immediate, zero-shot transfer.

D. Multi-Embodiment Robotics Control

The problem of creating a single policy for multiple robot morphologies is well-studied in legged robotics. Various strategies have been proposed to tackle this challenge, with recent efforts often leveraging architectural innovations to encode morphological information [29]–[31].

Unified Robot Morphology Architecture (URMA) [32] uses an attention mechanism to create a fixed-size latent representation from varied joint observations and descriptions, enabling control of different robot types.

Body Transformer (BoT) [33] represents the robot's body as a graph and uses masked attention to leverage the physical structure as an inductive bias. These approaches provide a strong conceptual foundation for our work.

We adapt and implement these two as powerful baselines, translating their discrete, joint-based representations to the continuous parameter space of vehicle physics.

E. Vehicle-Specific Dynamics Modeling

Recent works have begun to address vehicle dynamics more directly, though with different objectives than our own.

AnyCar to Anywhere [34] proposes a universal dynamics model for agile control, demonstrating impressive generalization for trajectory tracking tasks after fine-tuning on real-world data. However, its focus remains on dynamic prediction for agile mobility rather than learning a complete, reactive policy for complex urban driving scenarios.

Similarly, *One Model to Drift Them All* [35] successfully adapts a single model for extreme, at-the-limit maneuvers on two distinct real-world vehicles. Yet, its scope is intentionally limited to the specialized domain of drifting, not general on-road navigation, and it relies on online adaptation rather than achieving true zero-shot transfer to unseen vehicles.

Closer to our problem formulation, *Vehicle Type Specific Waypoint Generation* [36] presents a method to make a general behavioral model produce more physically plausible waypoints for a specific vehicle type. While it considers vehicle properties, it addresses the waypoint generation in certain turning scenarios only, not the creation of an integrated, end-to-end urban driving policy. These approaches highlight the importance of vehicle dynamics but also underscore the novelty of MVAdapt's goal: achieving robust zero-shot adaptation for a complete E2E driving policy in general traffic scenarios.

III. Methodology

The MVAdapt architecture (Fig. 3) is designed to integrate physical properties into an existing E2E driving model. It consists of a frozen pre-trained feature extractor and a physics-attention adaptation module.

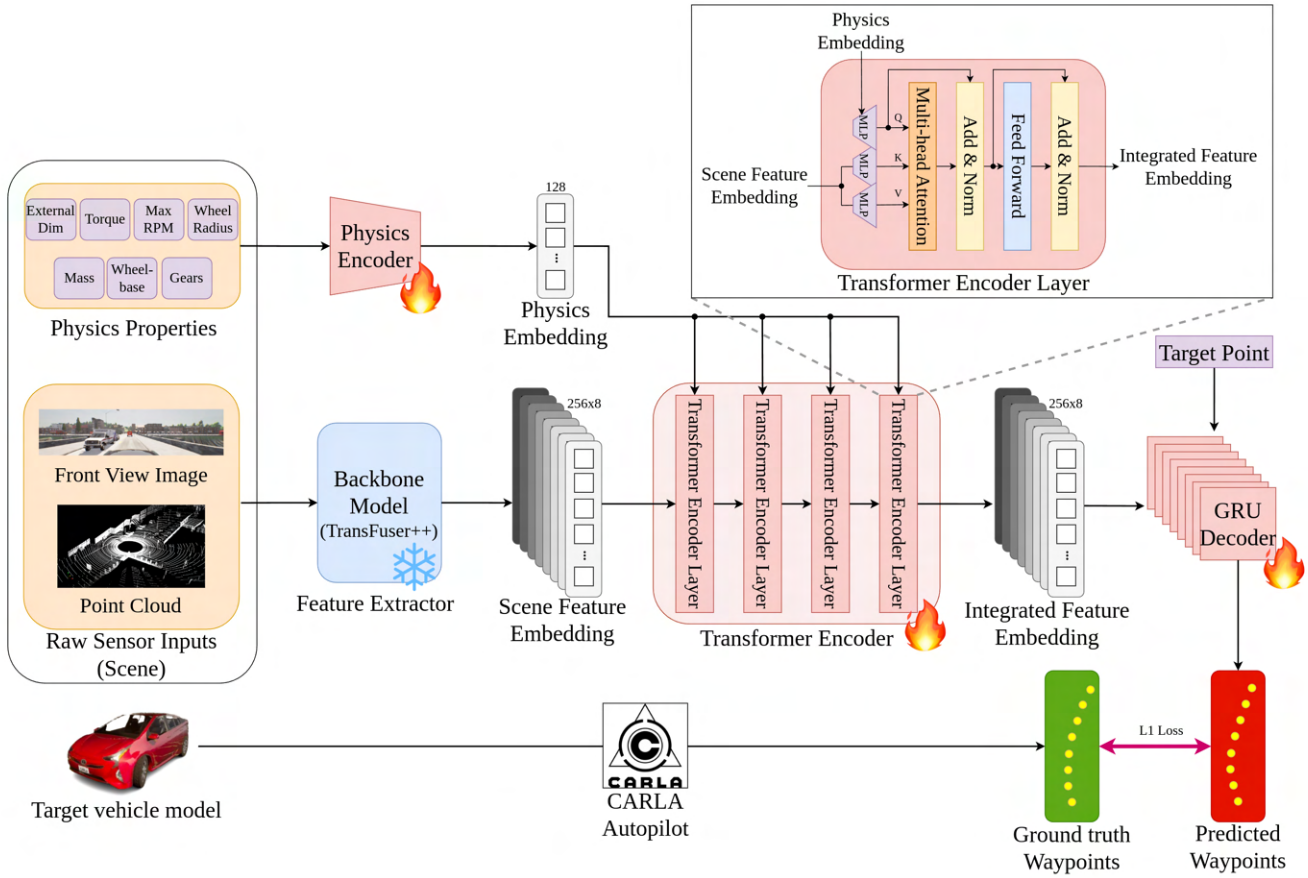

Fig. 3: *Overall architecture of MVAdapt*: Raw sensor inputs (camera image and LiDAR point cloud) are processed by a frozen TransFuser++ backbone to extract scene features, while vehicle-specific physical properties are encoded into a physics embedding. A multi-head transformer encoder fuses the physics embedding with scene features, producing an integrated feature embedding that conditions perception on the ego-vehicle's dynamics. Finally, a GRU-based decoder generates future waypoints toward the target point, trained with an L_1 loss against expert trajectories from CARLA autopilot.

A. Backbone Feature Extractor

To ensure robust multi-modal perception, we employ a well-structured E2E driving model, *TransFuser++ WP*, as our backbone [20]. TransFuser++ uses a multi-layered transformer architecture to fuse between two separate pipelines for image and LiDAR sensors. In our framework, we use the high-level part of this model as a feature extractor. Its weights are pre-trained on a dataset from CARLA and the `Lincoln MKZ 2017` vehicle model (default model). While the entire pipeline is trained, these weights are frozen. This transfer learning approach leads MVAdapt to build upon a guaranteed, general understanding of driving scenes, traffic, and semantics without the need for extensive training from scratch.

B. Physical Properties and Physics Encoder

We collect these multi-dimensional properties with the CARLA API (TABLE I), flatten them into a single vector, and apply zero-padding to ensure a uniform input size for our model. This processed vector is then passed through a small encoder network to generate a physics embedding. This embedding serves as a low-dimensional yet comprehensive representation of the vehicle's dynamic profile, encapsulating both its expected behavior *how the vehicle acts* and its operational limits *constraints to monitor*.

C. Multi-Head Transformer Encoder

We fuse scene representations with vehicle physics through a multi-head attention mechanism. Within the transformer encoder, the physics embedding *(Query)* attends to the scene feature embedding *(Key)*, allowing the model to dynamically focus on the most essential scene attributes at each time step. These attention weights are then applied to the scene feature embedding *(Value)*. This attention-driven fusion creates a physics-informed embedding for the target vehicle. In conclusion, the vehicle *(physics embedding)* asks the scene *(scene feature embedding)*, which scene attributes the model should attend to.

D. Output Head

The output from the adaptation module is conceptualized as an integrated scene feature embedding. This feature is subsequently fed into the final output head, which consists of a Gated Recurrent Unit (GRU). The GRU also takes the coordinates of the target point, which is sequentially given by the CARLA Leaderboard benchmark, as an additional input. It iterates through the stack of $N_{target} = 8$ scene feature embeddings to sequentially generate N_{target} time-variant waypoint coordinates. This sequence of waypoints constitutes the future trajectory of the ego vehicle. Finally, this predicted trajectory is passed to a fixed low-level PID controller, which generates the control inputs (i.e., throttle, steering, and brake).

IV. Experiments

A. Experimental Setup

1) Simulation Environment: We use the CARLA simulator [37] version 0.9.12, a standard tool for autonomous driving research. It provides an open-source platform built to support the development, training, and validation of autonomous systems. To standardize evaluation, the CARLA Autonomous Driving Leaderboard [38] is used. The benchmark challenges agents to navigate predefined routes across diverse environments while handling a variety of complex traffic scenarios.

2) Data Generation: With the default vehicle catalogue of the CARLA simulation, we can collect the ground truth driving datasets for 27 distinct vehicle models. For each vehicle archetype, a rule-based autopilot of the simulation generates driving datasets, which include sensor data, corresponding trajectory, and the physical properties of the vehicle. To ensure the quality of the supervising data, we selected the perfect driving scenarios without any accidents. Additionally, a huge truck and a tiny car are excluded to validate the adaptation ability to extreme unseen vehicles.

3) Benchmark and Metrics: We evaluate the driving performance of MVAdapt with all vehicles on the `longest6` benchmark, which consists of 6 routes in 6 different towns. Performance is measured using the official CARLA Leaderboard metrics:

- **Route Completion (RC)**: The percentage of the route that the agent completed before the termination of the scenario due to a high-risk accident (e.g., collision, rollover, off-road, stuck, etc.).

$$RC = \frac{l_{complete}}{l_{total}} \tag{1}$$

- **Infraction Score (IS)**: A penalty factor that decreases from 1.0 for every traffic violation (e.g., collision, ignoring a stop sign, traffic light violations, etc.)

$$IP = \prod_k p_k \tag{2}$$

$$p_k \sim (0,\ 1) \text{ on every violation}$$

$$p_k = \begin{cases} 0.5 & \text{Collisions with pedestrians} \\ 0.6 & \text{Collisions with other vehicles} \\ 0.65 & \text{Collisions with static elements} \\ 0.7 & \text{Running a red light} \\ 0.8 & \text{Running a stop sign} \end{cases}$$

- **Driving Score (DS)**: The primary metric, calculated as the product of Route Completion and Infraction Score.

$$DS = RC_i IP_i \tag{3}$$

B. Training

The model is trained on a dataset from 27 vehicles, which consists of about 1M frames total. This training procedure maps the multi-modal sensor data and the future waypoints. $L1$ loss is used between the predicted waypoints and the rule-based expert's waypoints.

C. Zero-Shot Adaptation

To analyze the zero-shot adaptation ability of the proposed method, we deployed the trained AI model to several unseen vehicle models. They contain not only the excluded vehicles from the CARLA catalogue, but also vehicles with stochastically distributed physics. The physical properties of these *'sampled vehicles'* are modified with the CARLA API, so that the vehicles act based on the sampled physics. To ensure realistic maneuvers, the sampling range for each physical parameter was carefully determined. This experimental setup has enabled a more diverse and flexible range of zero-shot adaptation experiments.

D. Few-Shot Adaptation with Fine-tuning

Additionally, we conduct a few-shot adaptation experiment: we fine-tune MVAdapt on a tiny dataset ($\sim$ 37K frames, $\approx$ 3% of the full training) of the `Carla Cola Truck` driving, then evaluate its performance on that truck. This simulates a scenario where a new vehicle becomes available, and we update the model with minimal data. We examine how quickly performance improves on the vehicle.

E. Baselines

We compare MVAdapt against three baselines:

- *Naive Transfer*: This is the case where we use the backbone model and its low-level decoder only (TransFuser++ waypoint model). It is trained on the source vehicle (`Lincoln MKZ 2017`) and deployed to the target vehicles without any adaptation. It represents the standard practice of training an end-to-end model per vehicle and reveals the drop in performance when the embodiment changes. This baseline helps quantify the performance gap due to vehicle mismatch.
- *URMA-style adapter*: To compare with prior adaptation methods, we implement a variant of *One Policy to Run Them All (URMA)* for the driving task. In URMA, a policy gets a description of the agent's body (e.g., joint parameters) and uses attention to condition the action generation. We adapt this idea by providing a learned encoding of the vehicle's properties to the policy and

TABLE I: Physical Properties for MVAdapt

Physical Property	**Dimension**	**Description**
External Dimension	3	The overall size of the vehicle in meters (length, width, height).
Torque Curve	$2 \times N_{\text{torque}}$	Data points representing engine torque (Nm) at various RPMs.
Max RPM	1	The maximum rotational speed the engine can achieve.
Mass	1	The total mass of the vehicle (kg).
Center of Mass	3	The (x, y, z) coordinates of the vehicle's center of gravity.
Wheelbase	1	The longitudinal distance between the front and rear axles.
Wheel Radius	1	The radius of the vehicle's wheels.
Forward Gears	$3 \times N_{\text{gears}}$	Defines each gear by its ratio, up-shift RPM ratio, and down-shift RPM ratio.

TABLE II: In-distribution Vehicles (Average over 27 vehicles)

Model	**Avg. Driving Score (DS) ↑**	**Avg. Route Completion (RC) ↑**	**Avg. Infraction Score (IS) ↑**
TransFuser++ (Default Vehicle)	80.27	96.33	0.83
TransFuser++ (Naive Transfer)	19.31	46.85	0.45
URMA	33.25	46.63	0.71
BodyTransformer	36.92	58.86	0.56
MVAdapt (Ours)	**78.21**	**96.62**	**0.80**

TABLE III: Out-of-distribution Vehicles (Average over 31 vehicles)

Model	**Avg. Driving Score (DS) ↑**	**Avg. Route Completion (RC) ↑**	**Avg. Infraction Score (IS) ↑**
TransFuser++ (Naive Transfer)	28.77	54.23	0.51
URMA	24.39	46.83	0.48
BodyTransformer	23.21	51.06	0.43
MVAdapt (Ours)	**63.02**	**96.77**	**0.65**

TABLE IV: Fine-Tuning on unseen `Carla Cola Truck`

Model	**Avg. Driving Score (DS) ↑**	**Avg. Route Completion (RC) ↑**	**Avg. Infraction Score (IS) ↑**
Performance on `Carla Cola Truck`			
MVAdapt (Unseen, before FT)	30.37	100.0	0.30
MVAdapt (Fine-tuned)	**61.90**	**100.0**	**0.62**

integrating it via a soft attention mechanism, similar to URMA's approach. Therefore, the URMA baseline has access to the same vehicle parameters as MVAdapt, but its architectural integration has simpler attention than our full transformer fusion. It also samples stochastic actions during training, as in the original URMA, for robustness.

- *BodyTransformer-style adapter*: We also adapt the *BodyTransformer* baseline to our setting. *BodyTransformer* introduced a way to integrate structured information about an agent's body as a graph of joints into a transformer policy. This baseline is also given the same vehicle properties as our method, but uses a different network design. The baseline treats the physical properties as a *body part list* and applies a transformer encoding similar to *BodyTransformer* to mix those. It provides another point of comparison for how to utilize embodiment information.

MVAdapt, URMA-style, and BodyTransformer-style models are trained with the same dataset to ensure an accurate comparison.

V. Results

A. Quantitative Performance

1) Performance on In-Distribution Vehicles: MVAdapt achieves sufficient driving performance on the vehicles it was trained on (in-distribution), using one unified model. TABLE II summarizes the results. MVAdapt achieves an average Driving Score of 78.21, outperforming the naive TransFuser++ transfer (DS 19.31) and significantly surpassing the URMA (DS 33.25) and BodyTransformer (DS 36.92) baselines. MVAdapt's Driving Score is comparable to TransFuser++, which is specialized and trained on a specific vehicle, achieving a score of over 80 on its own vehicle.

More impressively, MVAdapt attains an average Route Completion of 96.92%, meaning it can drive through the entire route without a termination, most of the time. In contrast, the baselines show 47-59% RC score on average, as they often fail to complete due to crucial accidents such as collisions or getting stuck. This route completion improvement is critical, as the proposed method can avoid high-risk mistakes. The IS of MVAdapt is 0.80, which is also the highest. It reflects fewer traffic rule violations. In summary, under in-distribution vehicles used in training, MVAdapt maintains robust, safe driving across vehicles, whereas the naive method or other adaptation approaches struggle with some vehicles.

2) Performance on Out-of-Distribution (Unseen) Vehicles (Zero-Shot Adaptation): On completely unseen vehicle types (out-of-distribution test), MVAdapt still delivers strong performances, demonstrating successful zero-shot adaptation. As shown in TABLE III, MVAdapt's average DS is 63.02 when facing novel vehicles. This is notably higher than the naive transfer (DS 28.77) and also the URMA and BodyTransformer baselines (DS 24.39 and 23.21, respectively). In other words, without any retraining, MVAdapt retains a high level of driving competence on vehicles it has never encountered before, while other methods show poor performances. The RC remains notably high at 96.77%. This proves that MVAdapt can drive new vehicles without causing catastrophic failures by understanding their physics and dynamics. It also carries out a better IS. These results validate our design: by leveraging the vehicle's physics parameters, MVAdapt can immediately adjust its driving policy to accommodate differences like a bigger external dimension or wider turning radius in a new vehicle, without needing new training data. To our knowledge, this is the first demonstration of such zero-shot cross-vehicle generalization in an end-to-end urban autonomous driving task.

Notably, MVAdapt's in-distribution performance is close to a model optimized for a specific car (TransFuser++ on its default vehicle scored DS 80.27, RC 96.33%). And although performance drops for unseen vehicles, MVAdapt still achieves DS over 60 with RC over 97%.

3) Performance Improvement through Few-Shot Adaptation: For the exceptionally challenging case of the Carla Cola Truck (a physical outlier), we found that a few fine-tuning data can significantly improve MVAdapt's performance on this extreme vehicle. Before fine-tuning, the zero-shot DS on the truck under MVAdapt is about 30.37, much lower than the other zero-shot adaptation scores, as expected for a vehicle with drastically different physics. The truck could complete all of the routes as it shows 100% of RC. However, it comes with many infractions (IS 0.30), likely due to collisions or blocking in narrow spaces. After fine-tuning MVAdapt on a small dataset of truck driving, the DS jumped to 61.9, and the IS improved to 0.62. This bridged most of the performance gap, proving that MVAdapt can rapidly adapt to even an outlier vehicle in such a few-shot adaptation manner.

This experiment highlights a practical deployment scenario: MVAdapt could be shipped as a general model, and if a new vehicle type, especially one far from the training distribution, needs better performance, a small calibration run on that vehicle can quickly boost performance without needing to recollect a full dataset or train a new model from scratch.

B. Qualitative Analysis

Qualitative observations from simulation runs further illustrate how MVAdapt adapts driving behavior to target vehicles. In various test routes, baseline agents often encountered

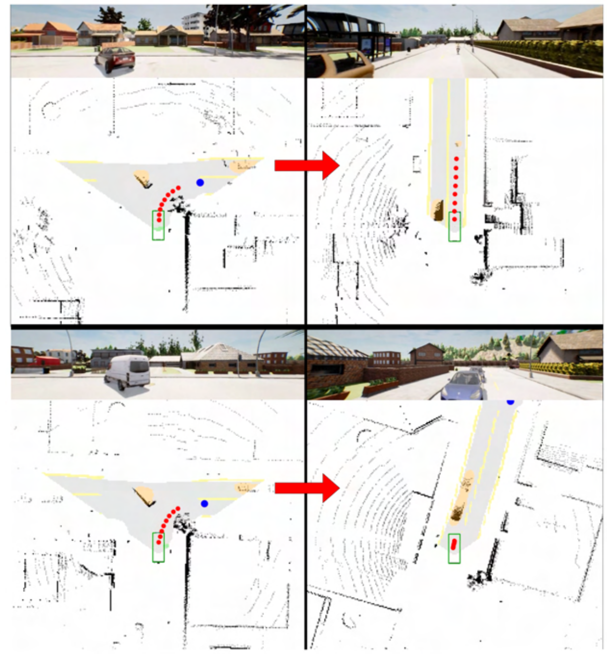

Fig. 4: A `Tesla Cybertruck` making a right turn. **Top (Ours)**: MVAdapt successfully navigates the turn by accounting for the vehicle's large size. **Bottom (Baseline)**: The baseline misjudges the turning radius and gets blocked by another car. Red dots are the model's output trajectory, and the blue dot is the target point.

problems that MVAdapt handled correctly:

1) Maneuver Feasibility (Turning Radius): In one scenario with a large vehicle (`Tesla Cybertruck`), the URMA-based agent misjudged the turning radius and was blocked by another car, resulting in a scenario termination (Fig. 4). In contrast, MVAdapt, aware of the Cybertruck's external dimension via its physics embedding, maintained a safer trajectory so that it could pass through in the identical situation. This led to MVAdapt completing the route successfully (RC 100%), whereas URMA was prematurely terminated (RC $\sim$66%).

In another case with a very tiny car (`Mini Cooper`), the baseline BodyTransformer agent attempted a right turn at an intersection at a trajectory curvature appropriate for a midsize car: however, the Mini Cooper's tighter turning ability meant it over-rotated and ended up partially off-road, triggering an infraction and getting stuck on the curb (Fig. 5). MVAdapt, by considering the vehicle's small size and agile steering, executed a smoother, tighter turn without leaving the lane. The outcome was MVAdapt completing the turn and route (RC 100%) with no infraction, whereas the baseline failed the route (RC 15%). This suggests MVAdapt's knowledge about the minimum turning radius constraint and vehicles' turning reactions: it learned to adjust planned paths to ensure they are physically achievable by the specific vehicle.

2) Out-of-Bounds Predictions: We also noticed that the baseline models sometimes generate out-of-bounds predictions in certain situations. Fig. 6 presents the baseline

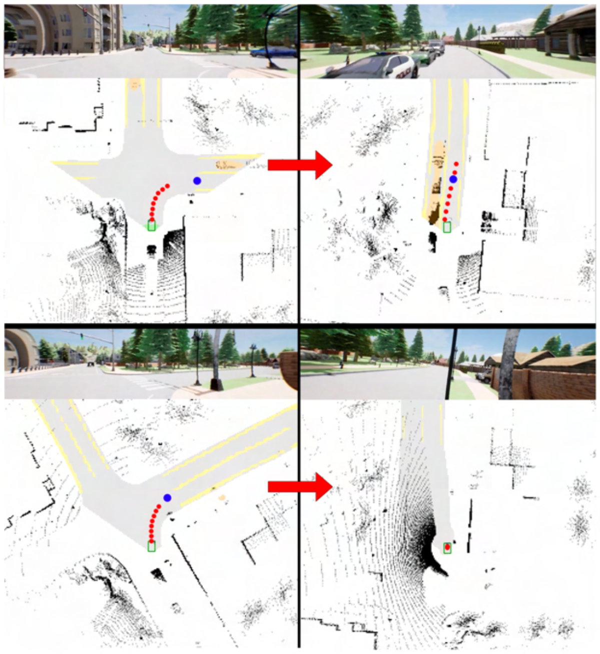

Fig. 5: A `Mini Cooper` turning right. **Top (Ours)**: MVAdapt executes a smooth, tight turn appropriate for the vehicle. **Bottom (Baseline)**: The baseline model over-rotates and hits the curb, failing the maneuver. Red dots are the model's output trajectory, and the blue dot is the target point.

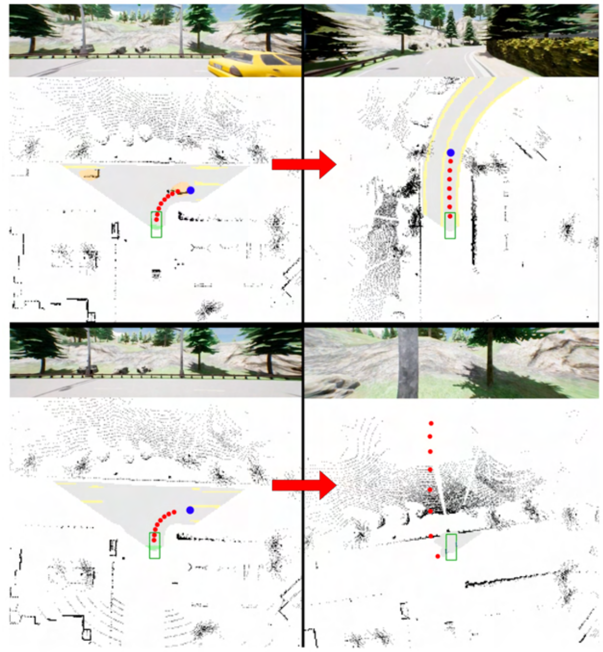

Fig. 6: Catastrophic failure during a right turn. **Top (Ours)**: MVAdapt generates a stable trajectory and safely completes the turn. **Bottom (Baseline)**: The baseline predicts an out-of-bounds trajectory, leading to an immediate failure. Red dots are the model's output trajectory, and the blue dot is the target point.

BodyTransformer model suffering from a catastrophic failure during a right-turn maneuver. The BoT policy generated a trajectory that veered out of bounds mid-turn, leading to an immediate scenario termination. In contrast, MVAdapt executed the same tight right turn safely within lane boundaries. This case highlights MVAdapt's ability to handle complex maneuvers for vehicles with different physical characteristics. The proposed method demonstrates the most accurate mapping between the physical properties of the target vehicle and the corresponding driving policy, leading to stable trajectories.

VI. Conclusion

We presented *MVAdapt*, a universal end-to-end autonomous driving model capable of adapting to multiple vehicle types, including vehicles not seen during training. By integrating vehicle physical properties into the decision-making through a transformer-based attention module, MVAdapt can generalize a learned driving policy across a spectrum of vehicle dynamics. It achieves state-of-the-art performance on challenging urban driving tasks in CARLA, outperforming on both in-distribution and zero-shot transfer scenarios. Our experiments demonstrated that MVAdapt maintains high route completion and low infractions across vehicles. It significantly closes the performance gap that occurs when deploying a driving model on an unseen vehicle, with minimal fine-tuning. This fine-tuning ability highlights its practicality for real-world usage, where new vehicle models may be introduced over time.

For future work, we plan to extend MVAdapt in several directions. First, we will explore adaptation to not just vehicle parameters but also *hardware and actuator differences* (steering dynamics, latency, etc.), making the policy robust to a wider range of embodied differences. Second, testing on real-world driving data or actual vehicles will be important to validate that the simulation findings carry over. To do so, several path planning algorithms [39]–[41] might be needed to provide target points. Third, implementing online adaptation is a promising future research topic. Similar to how a person adapts to a new vehicle, online adaptation during driving could be an effective method for achieving multi-vehicle adaptation. We hope that our approach inspires more research into driving *foundation models* that are generalizable across many vehicles while specialized in exploiting each vehicle's unique physics.

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